

Dilations: Building and Measuring Similar Figures

Dilating points on the coordinate plane, finding a scale factor from two lengths, and using a scale factor on a similar figure. Units are stated in every problem.

- A dilation with centre C and scale factor k sends each point P to the point P' on ray CP with $CP' = k \cdot CP$. Centred at the origin, $(x, y) \mapsto (kx, ky)$; centred at (h, j) , $(x, y) \mapsto (h + k(x - h), j + k(y - j))$.
- A dilation image is *similar* to its pre-image: every length is multiplied by k , every angle is unchanged, and area is multiplied by k^2 .
- Give lengths in the units of the problem and show the computation that produced each one.

1. A slide projector maps the slide onto a screen by a dilation centred at the origin with scale factor 3; coordinates on both the slide and the screen are measured in centimetres. The point $A(2, 1)$ is printed on the slide. Give the coordinates of its image A' on the screen.

Answer: _____

2. A triangular sail is cut from a pattern with vertices $A(0, 0)$, $B(4, 0)$ and $C(4, 3)$, measured in metres. The full-size sail is the image of that pattern under a dilation centred at $A(0, 0)$ with scale factor 2.5. Find the length of the full-size sail's side $B'C'$, then the length of its side $A'C'$.

Answer: _____

3. On a blueprint a rectangular patio is drawn 6 centimetres long and 4 centimetres wide. A print shop enlarges the blueprint by a dilation so that the long side of the drawing measures 21 centimetres. Find the scale factor of the enlargement, then find the width of the enlarged drawing.

Answer: _____

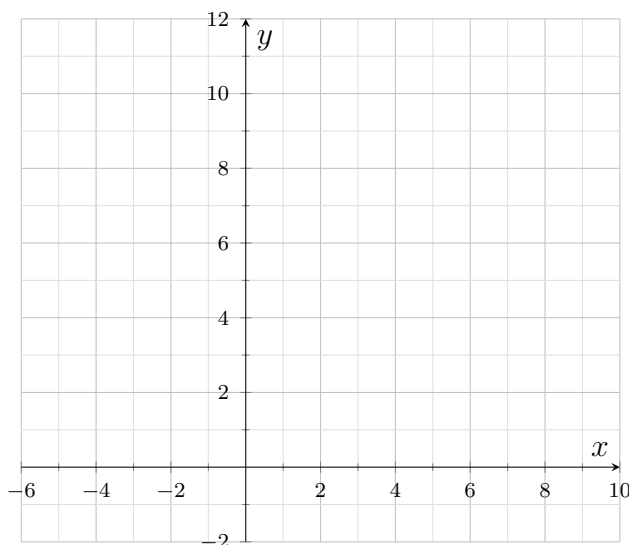
4. A digital map zooms about the pin at $P(1, 2)$ with scale factor 3; coordinates are in kilometres east and north of the town hall. A trailhead sits at $Q(3, 5)$. Give the coordinates of Q' , the trailhead's position on the zoomed map.

Answer: _____

5. A photograph measures 10 centimetres by 15 centimetres. A print shop enlarges it by a dilation with scale factor 2.4. **(a)** Find the two side lengths of the enlargement. **(b)** The enlargement's area is how many times the original's area?

Answer: _____

6. Triangle DEF has vertices $D(-2, 1)$, $E(4, 1)$ and $F(4, 5)$, with coordinates in metres. It is dilated about the origin with scale factor 2. **(a)** Plot DEF and its image $D'E'F'$ on the grid. **(b)** Compute the length DE and the length $D'E'$. **(c)** Divide the image length by the pre-image length and say what that quotient is.

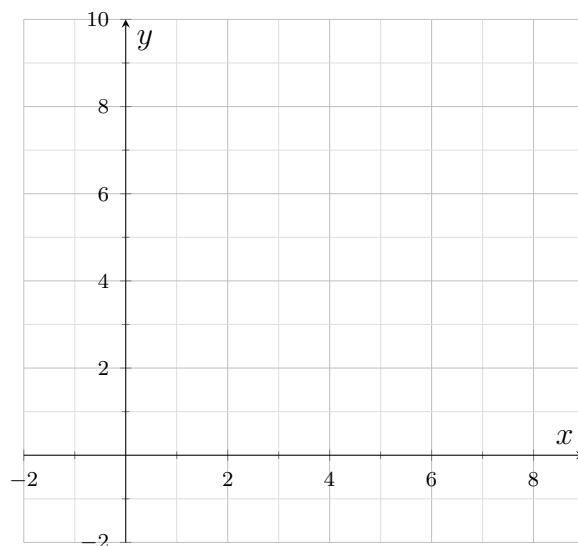


Answer: _____

7. A display model of a car is built as a dilation of the real car with scale factor $\frac{1}{12}$. The real car is 4.8 metres long, and the glass in its windshield has area 0.72 square metres. Ben computes the model's windshield glass area as 0.06 square metres. Explain what Ben did wrong, then give the correct area of the model's windshield glass in square metres, and the model's length in metres.

Answer: _____

- 8.** Triangle JKL has vertices $J(1, 1)$, $K(3, 1)$ and $L(3, 4)$, with coordinates in units. **(a)** Plot JKL and its image $J'K'L'$ under the dilation centred at the origin with scale factor 2. **(b)** Find the length $K'L'$. **(c)** Draw the line through J and J' , the line through K and K' , and the line through L and L' . Describe what those three lines do, and explain why triangle $J'K'L'$ must be similar to triangle JKL .



Answer: _____